From source documents to transaction decisions

A prospectus can permit a particular purchaser without establishing that a bank may execute the purchase. The bank must also establish which laws apply, who owns the account, whether sanctions restrict the transaction, and what its own mandate and settlement arrangements permit. LegalMath keeps these questions separate so that an answer to one cannot silently answer the others.

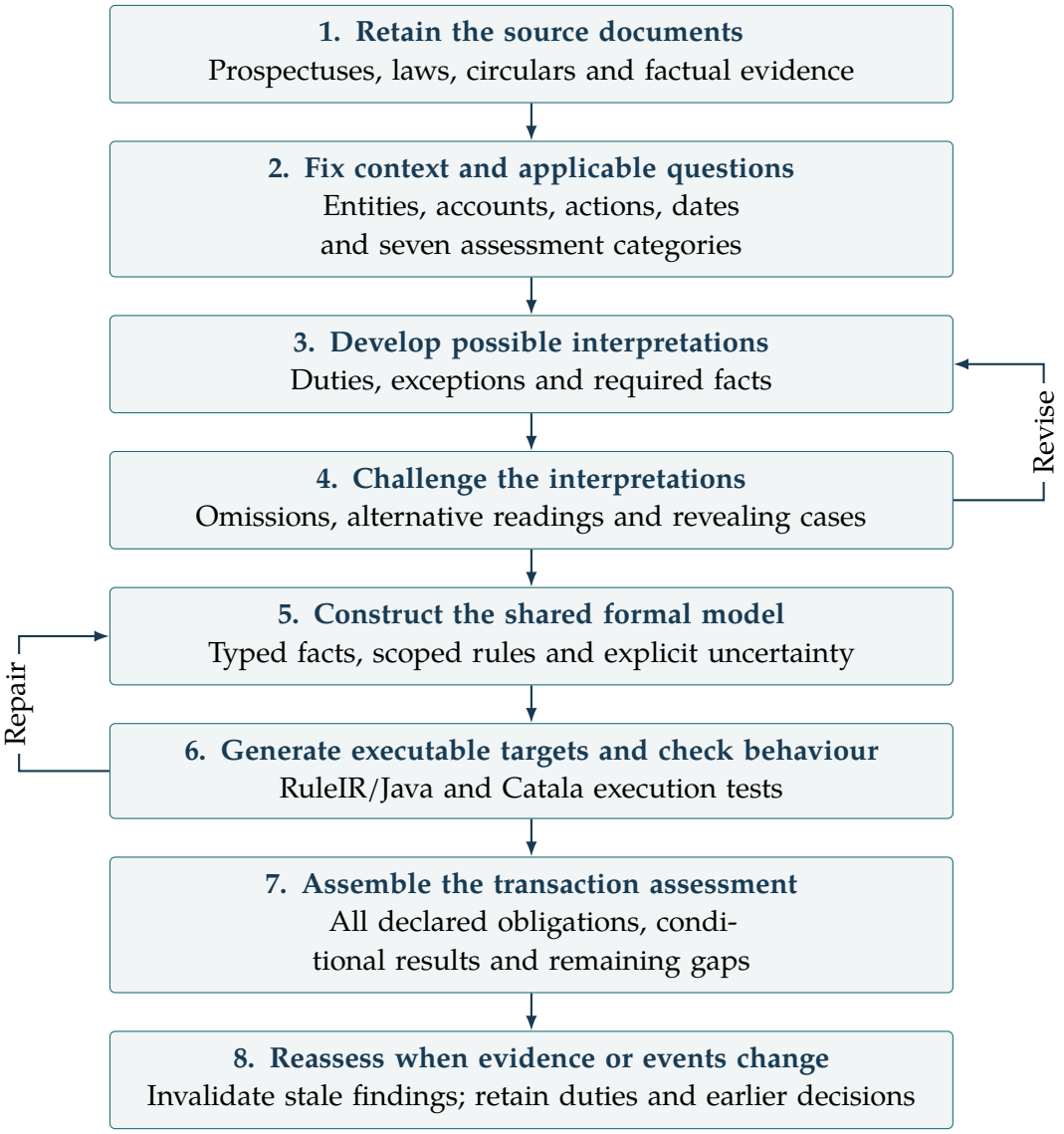
The running example compares two real contingent convertible securities, or *CoCos*: Barclays' February 2025 issue and Standard Chartered's January 2025 issue. These securities can absorb losses through conversion into shares and other contractual or regulatory measures. The same hypothetical US individual passes one selected purchaser condition and fails the other. Both complete bank assessments remain qualified because essential evidence and legal coverage are missing. The following charts trace that distinction.

There are two kinds of layer to keep apart. The *software stages* turn source documents and proposed meanings into executable calculations. The *seven assessment categories* organise the different legal and factual questions about a transaction. Every category uses evidence and scoped rules; it is not a separate programming language. The eight-step overview links to both the worked CoCo case and the implementation details.

The first software stage, called the *frontend*, proposes a meaning for the passage and records its sources, assumptions and required facts. A *translator* then turns that fixed proposal into an executable language. LegalMath has two such targets. *RuleIR*, its own small rule language, expresses conditions such as a financial threshold and the effect of missing evidence. *Catala* is a programming language designed for legal rules, with a compiler that translates those rules into executable code. Both targets can receive the same proposed meaning; Catala also supports the implemented calculations over structured collections described in Chapter 8.

Stipula answers a different question: how does a contract change when a party takes an action? It describes parties, resources and states. For example, a request available while an agreement is active may become unavailable after withdrawal. The book studies Stipula's verification methods as a comparison; LegalMath does not currently compile Stipula contracts.

Eight steps from documents to a qualified decision

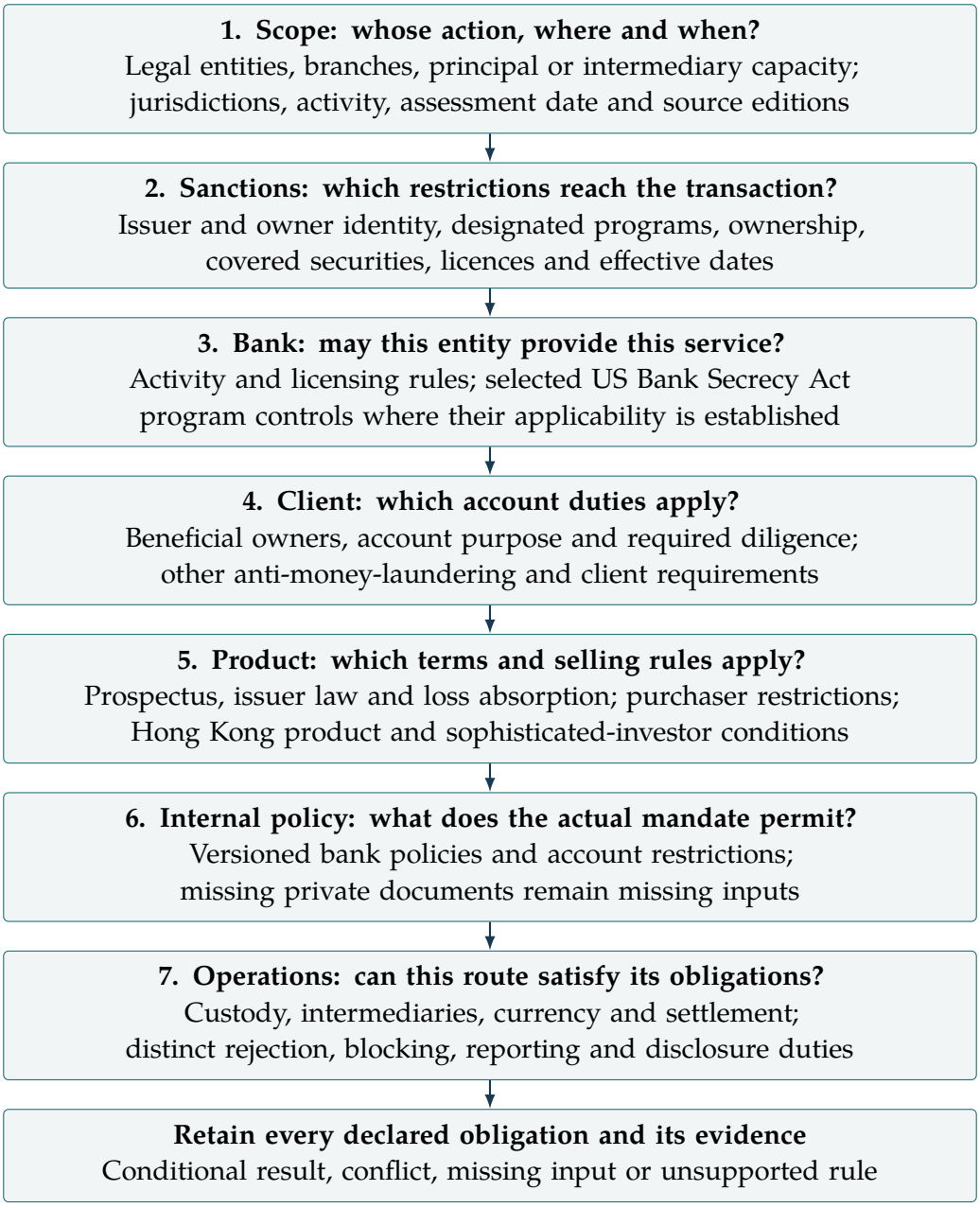


A failure or an unknown remains part of the assessment. Completing the software steps does not establish complete legal coverage or permission to execute a transaction.

The next pages expand the seven assessment categories and follow the two CoCo purchaser conditions into the complete bank assessment. Later pages explain each software step. Unsupported expressions prevent translation; missing facts can prevent execution even when a program has compiled. A different proposed interpretation remains a different model, with its own evidence.

The seven layers of a transaction assessment

Begin with a proposed action by a specified entity for a specified account. The bank's brand does not identify its booking branch, capacity or the client's status. Follow all seven categories: an adverse or unresolved finding remains in the assessment while the other questions are considered.

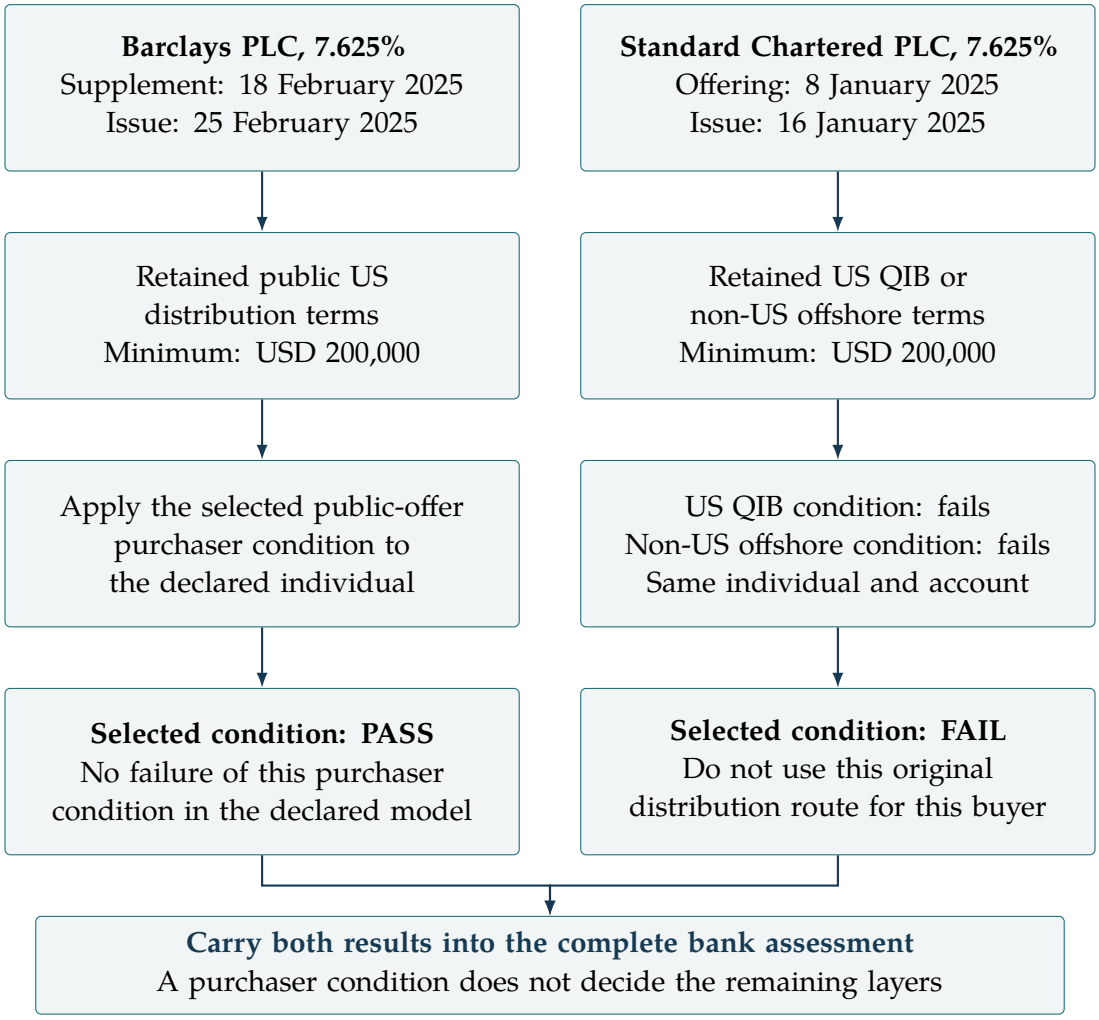


The current inventory has fourteen obligations across these seven categories. Five have conditional rule models; nine retain missing rule families or unresolved coverage. None may be removed by the caller.

The five calculated obligations concern selected sanctions, bank program controls, private-account diligence, sophisticated professional investor conditions and reporting. Supporting calculations bring the collection to eleven shared models. Complete discovery of applicable law remains unproved. Section 2.4 explains the provisions and their limits.

The CoCo example: from terms to purchaser conditions

Assume one individual, resident and located in the United States, buys USD 200,000 principal for their own account and benefit. The individual is not a *qualified institutional buyer* (QIB), an institutional and account category used by Rule 144A. Wealth alone does not turn an individual into a QIB. Regulation S supplies a different offshore distribution route; its US-person and account definitions must also be considered.



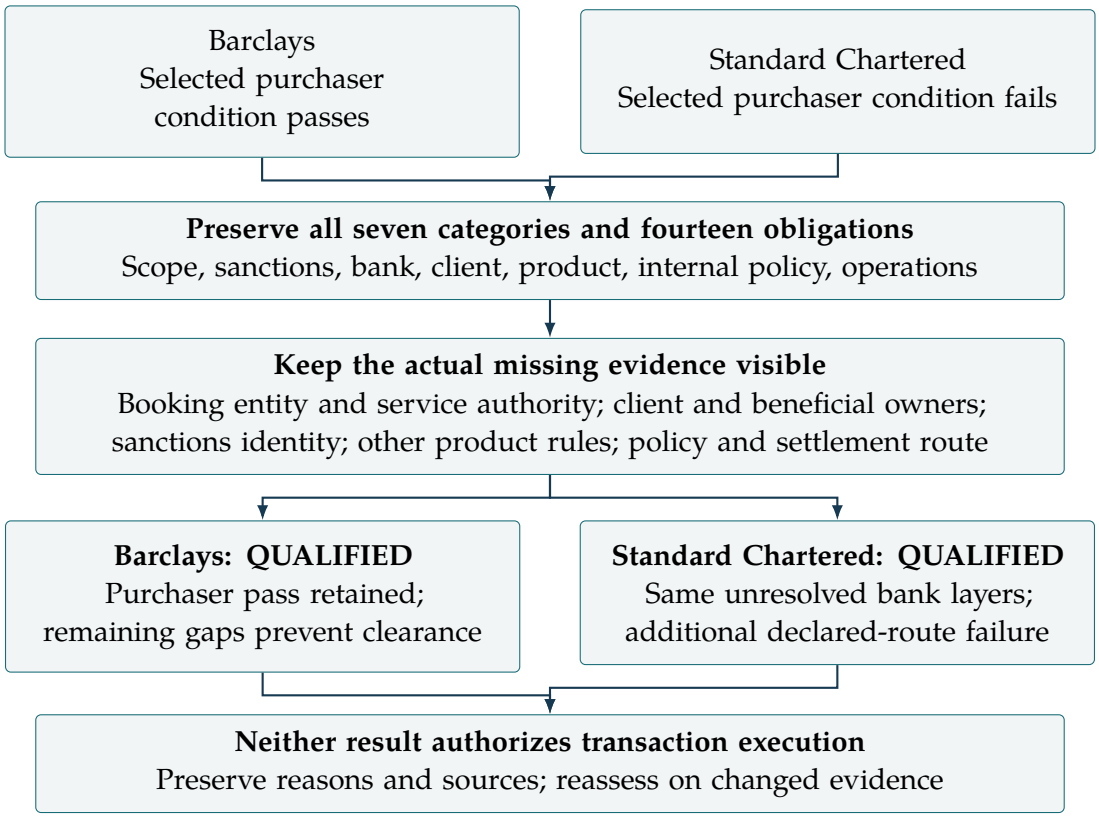
Both are real original offerings with the same minimum denomination. The difference is the selected purchaser restriction. Their offer dates differ; the comparison does not assert simultaneous live availability.

Both instruments can expose holders to conversion and regulatory loss absorption. A favourable purchaser result therefore says nothing about principal protection. The two prospectuses and the retained 2025 legal definitions support explicit interpretation proposals; page and file hashes identify that evidence without proving the proposals correct.

Changing the actual purchaser can change the answer. A QIB buying for itself can satisfy Standard Chartered's selected condition. A QIB bank acting for a non-QIB client cannot use its own status to satisfy that client-account condition. Moving an order offshore does not by itself remove US-person or US-benefit restrictions. Section 2.4.5 derives the formal rule and identifies the exact instruments and source passages.

The CoCo example: the complete bank result

The conditional purchaser finding supplements the existing bank investigation; it does not close the broader product-law obligation. A *qualified* result explains why the evidence cannot support a complete transaction decision.



The implemented attachment preserves the bank investigation and adds the purchaser finding. It cannot turn a partial pass into bank or regulatory approval. The negative result concerns the declared original distribution.

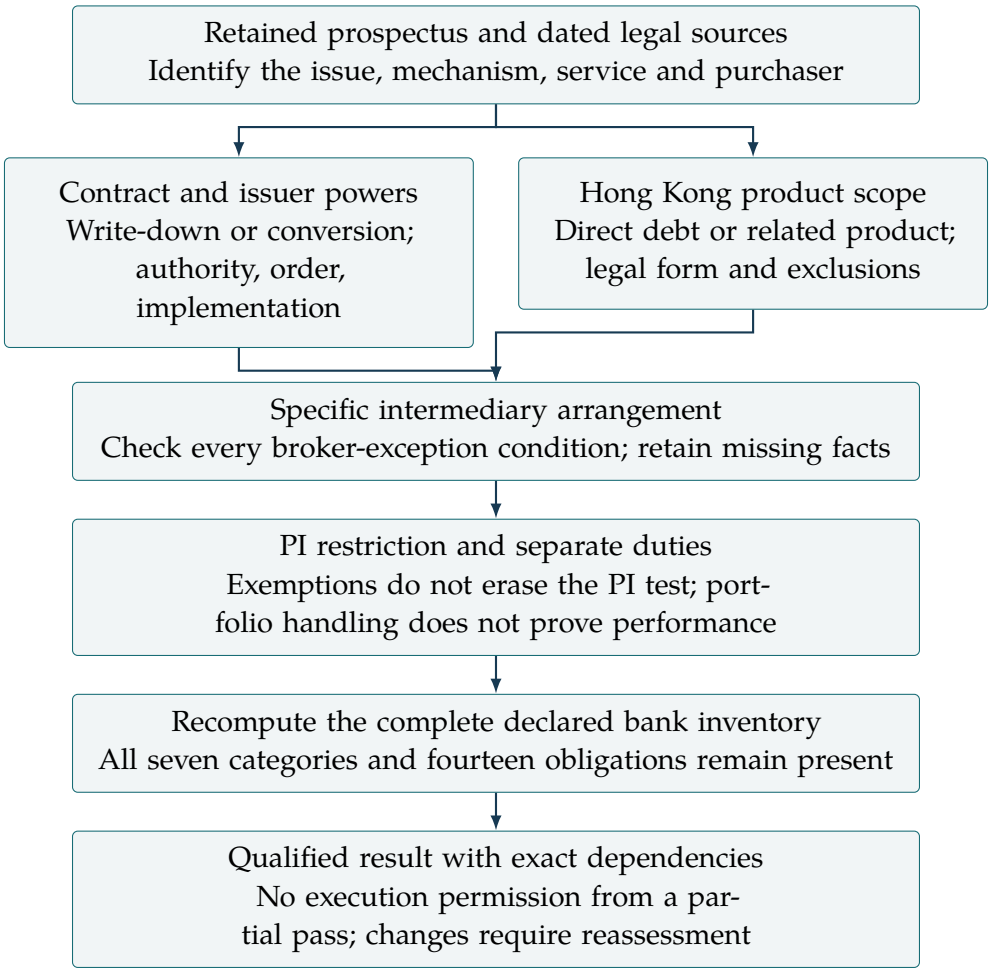
Issuer law is part of the product investigation even when the prospectus contains no decisive contractual trigger. The CoCo survey in Chapter 2 follows Swiss regulatory write-down powers and their dated legal and litigation context. That survey informs required source checks; it is not a completed executable model of every issuer jurisdiction. Swiss provisions cannot be applied to these UK issues merely because all are CoCos.

Sanctions require their own analysis of the issuer, beneficial ownership, program, activity and date. Chinese state ownership alone is not the selected sanctions trigger. A name match supplies a candidate identity; no match does not supply clearance. Restrictions on a US bank acting for a client also need the bank's actual capacity and the client's status. These questions remain separate from the US offering categories in the preceding chart.

The paired run reuses retained interpretation proposals; it does not conduct a fresh model investigation of every prospectus clause. The formal and execution checks support the stated calculation without proving its English interpretation or supplying missing bank evidence.

Join the product conditions to the bank investigation

The joined calculation keeps contractual loss, issuer powers and selling requirements distinct. Section 2.4.6 explains the implemented conditions and their remaining premises.

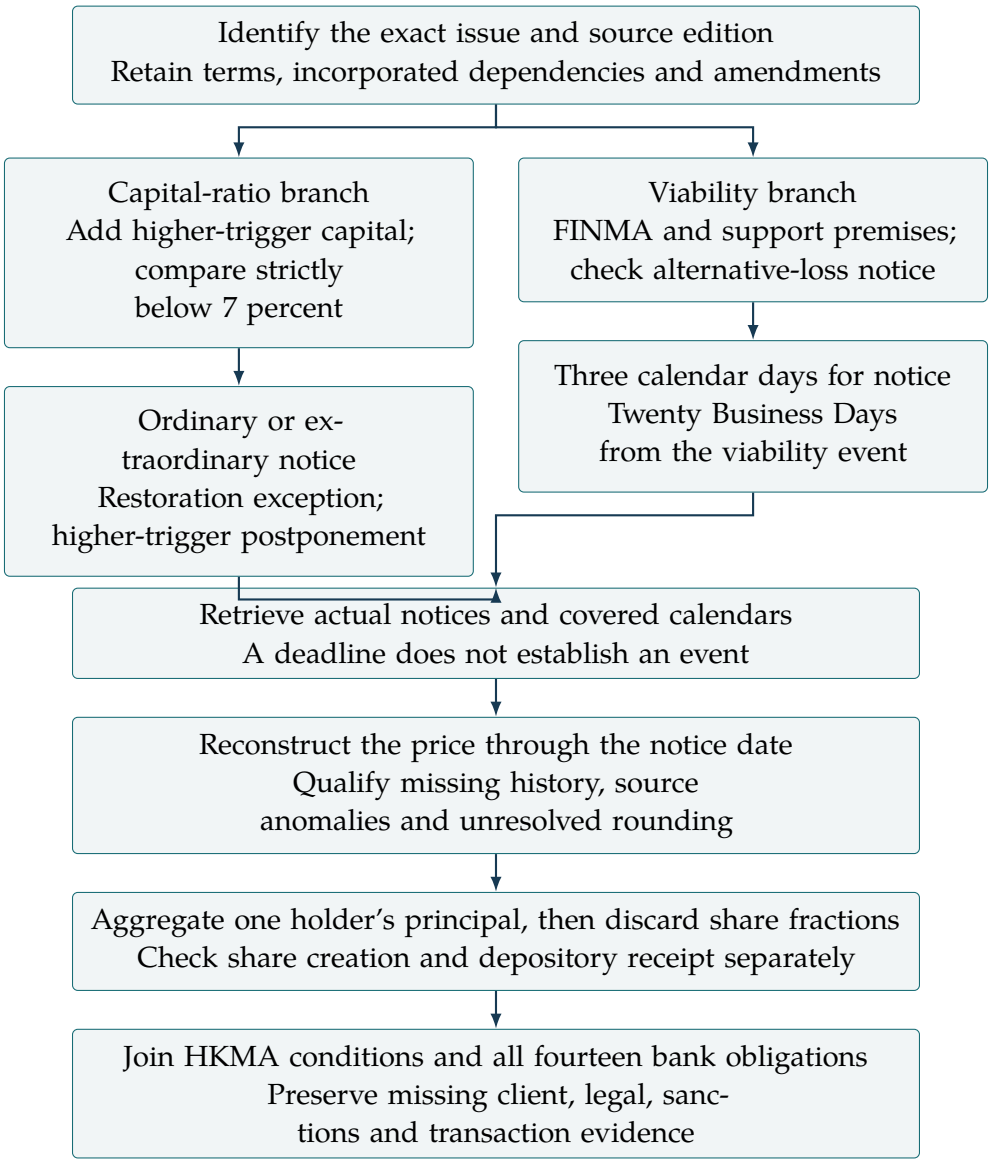


A conversion can cancel debt principal while delivering shares. Neither zero debt principal nor a PI classification settles the entire bank decision.

The retained UBS issue converts under its specified conditions. The next chart follows its implemented notice and share calculations. Swiss regulatory provisions do not establish powers over the UK issues. The actual account, mandate, bank entities and transaction route also remain necessary evidence. The joined reports for all three prospectuses retain these gaps.

Follow one UBS issue from publication to shares

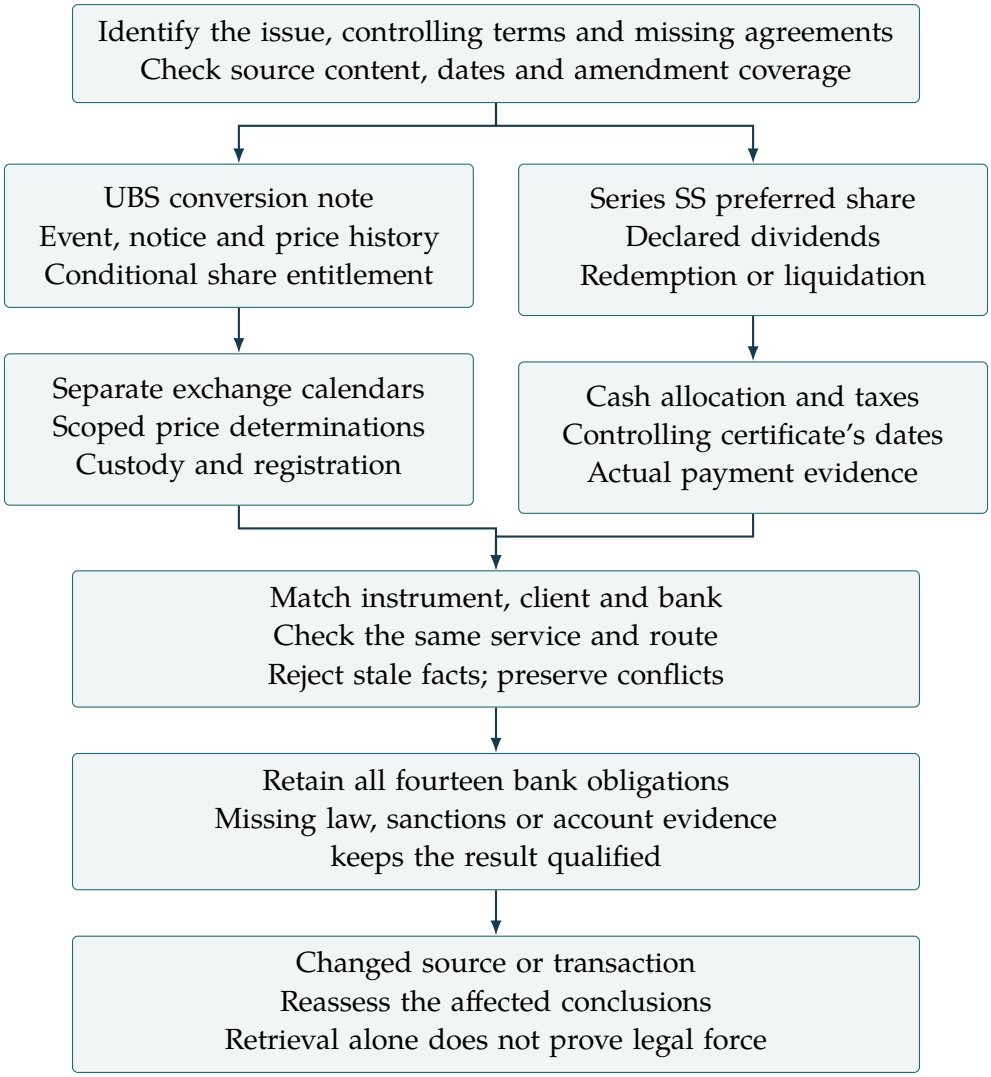
Section 2.4.7 uses Annex A of the June 2024 SGD notes. Source proposals populate the product analysis; actual client and event facts remain separate inputs.



The printed extraordinary-distribution formula has an unresolved denominator anomaly. Its literal calculation is retained; no replacement price or bank clearance is inferred.

Compare conversion notes with preferred shares

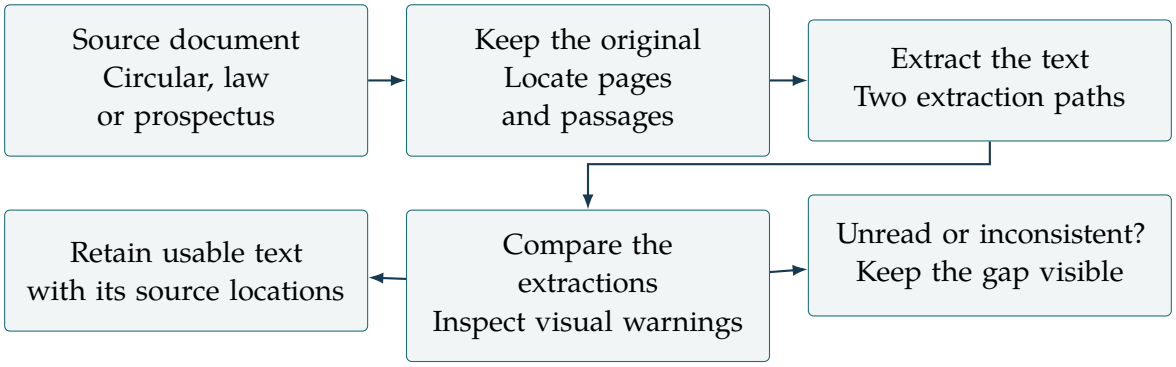
Section 2.4.8 follows the Series SS preferred share. Its non-cumulative dividend and redemption terms require different calculations from the UBS conversion note. Both still enter the same bank assessment.



Different instruments use different contractual calculations. Evidence about a distribution, conversion or delivery remains separate from permission for the bank to carry out the transaction.

1. Retain the source documents

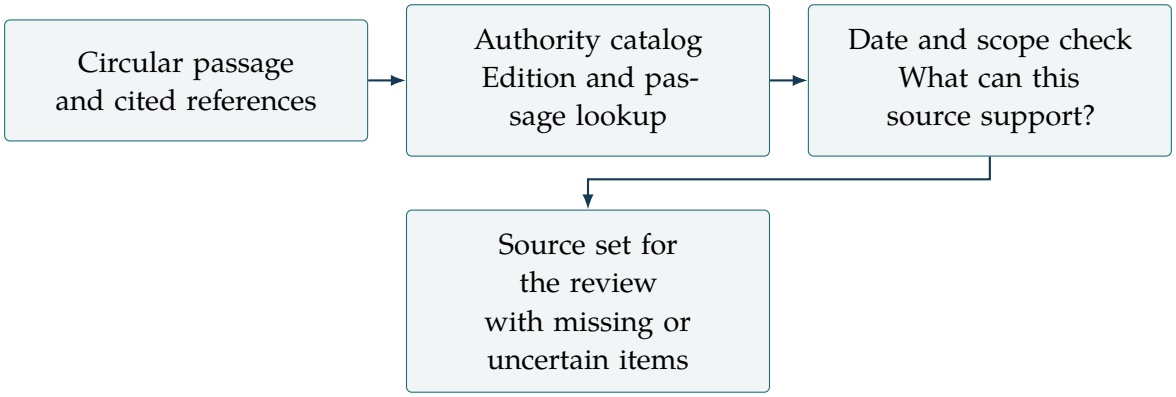
The first task is to give every later reading the same inspectable source. For PDFs, `pypdf` and Poppler's `pdftotext` are two different text extractors. Web pages have two parsing paths. Comparing their text can expose a missing line; checking the page for images can expose material that neither extractor has read.



Agreement between text extractors cannot reveal an exception present only in an image. The additional source checks in Chapter 6 read page images and compare the recognised text with the embedded text.

2. Fix context and applicable questions

A gift restriction may refer to the Code of Conduct and a frequently asked question (FAQ). A retained authority catalog connects those references to document editions and passages. A dependency search follows explicit official links; missing documents remain named questions. The relevant business scope identifies the people, products, activities and dates being assessed.

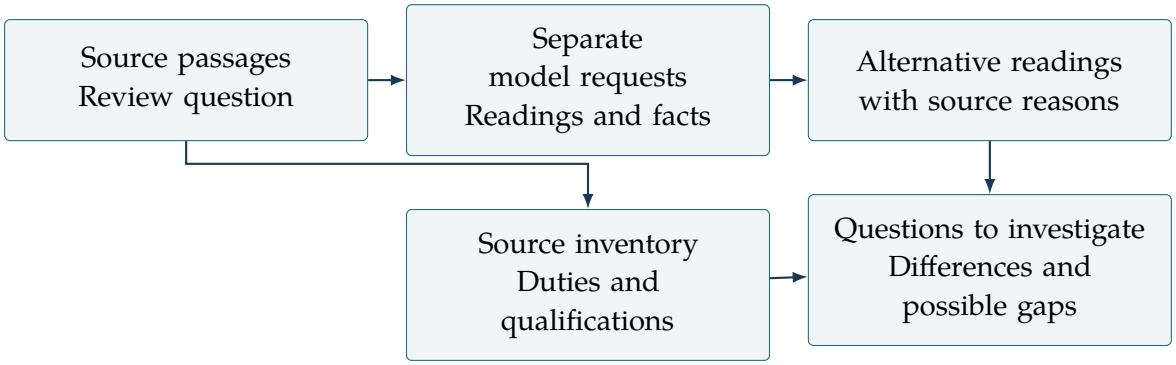


Finding the FAQ changes the question from where the source is to what its dated answer supports.

The retained FAQ page has later answer updates. Its older page date therefore does not establish everything a historical reader could have seen. Prospectus work adds issue-specific supplements, incorporated documents, issuer law, regulatory notices and the applicable selling rules. The transaction registry binds factual documents, policy and settlement-route records to their retained bytes and declared dates. Acquisition time does not establish legal force. Missing or conflicting evidence remains explicit. Chapter 2 develops these source questions; Appendix A gives the source tools.

3. Develop possible interpretations

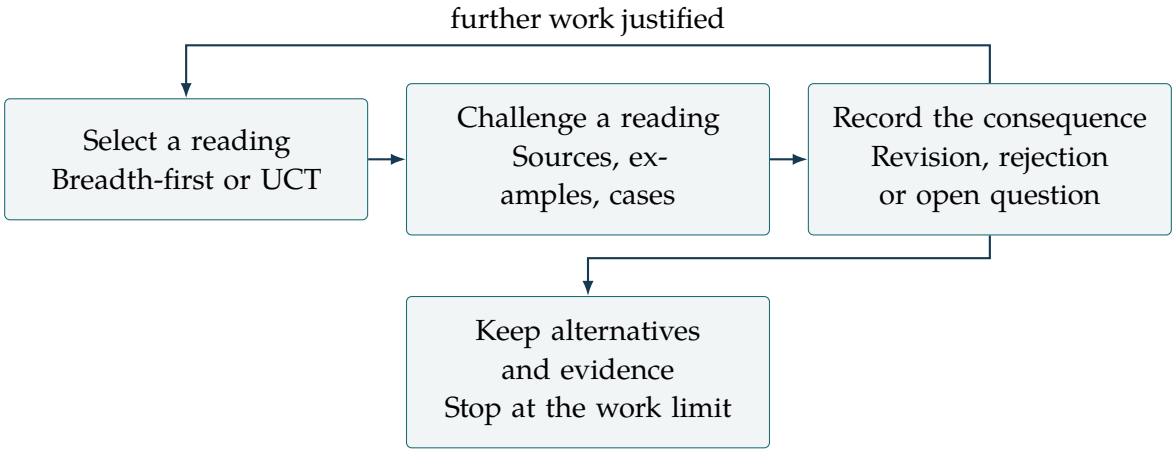
A language model can propose what a passage requires and which facts would decide a case. Separate requests produce initial readings without showing each request the other answers. Another set of requests inventories the source's requirements and qualifications. This second view helps expose an exception that all the proposed readings might miss.



Several readings reveal disagreement. A separate source inventory can reveal a shared omission. Separate requests can still share mistakes.

4. Challenge the interpretations

Tree search organises successive revisions of a reading. Breadth-first search visits the shallower alternatives first. Upper Confidence bounds applied to Trees (UCT) balances previously useful branches against less-visited ones. These are ways to choose the next investigation, not probabilities of legal correctness. Both are implemented with finite work limits.

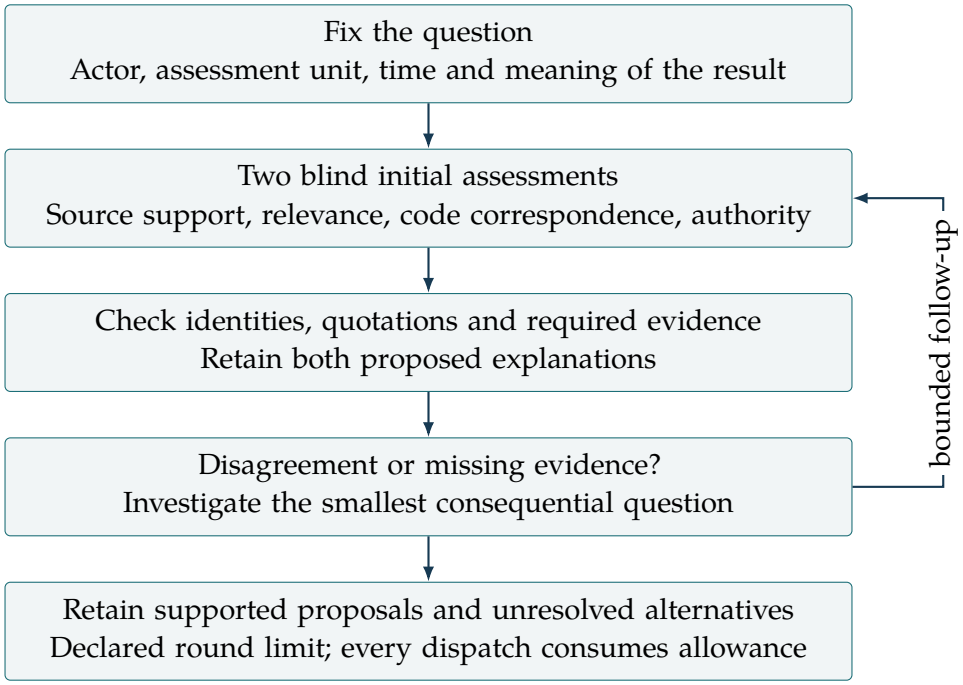


Search changes a proposed reading only with an explicit reason. Exhausting the work allowance leaves an unresolved result.

Typed arguments record support and objections; official examples retain their stated meaning. A rebate-and-voucher case can reveal where readings attach an exception. After aligning facts and outputs, the original programs execute that case. Draft compilation and step 6 therefore occur inside this loop too. Agreement still cannot establish complete source coverage. Chapters 5 and 6 develop these checks.

4 continued. Investigate the exact disagreement

The continuing investigation separates source support, relevance to the selected question, correspondence with the executable rule and availability of required authority. Two initial contexts work without peer answers. A later round can then address the precise disagreement, or repair an invalid response, within the same fixed allowance.



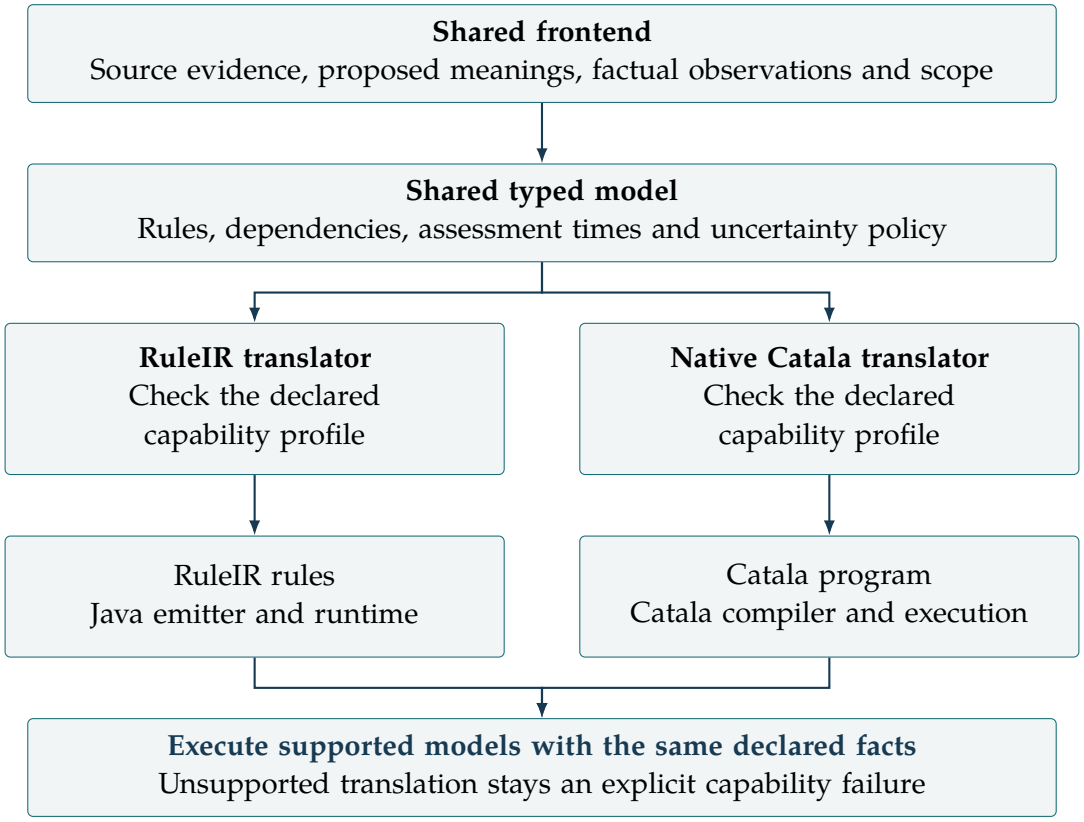
A repaired response is another proposal. Agreement cannot supply a missing source or prove an interpretation.

Rival factual models retain different actors, objects and assessment times. A proposed mapping permits a conditional comparison only after those distinctions and its assumptions are stated. Changed source bytes or methods start a new revision; the old program remains evidence for its old source. Section 6.35 follows the implemented distinctions through a contact history for a suspicious transaction report and the unfamiliar-circular study.

Capacity is checked separately from meaning. Smaller source or comparison batches retain the complete set of obligations and the original retry counts. The optional incremental check admits each validated part only while total retained size remains within its limit. A failed repair preserves earlier claims; it cannot replace them with an empty inventory. If the work still cannot finish, its missing checks remain in the final report.

5. Construct one shared model before choosing a target

A program needs the meaning of each fact and operation fixed. The shared frontend records sources, proposed interpretations, typed facts, scoped rules and uncertainty before a target language is selected. Type and dependency checks expose invalid combinations and missing or circular references. Separate translators then report which parts they can express.

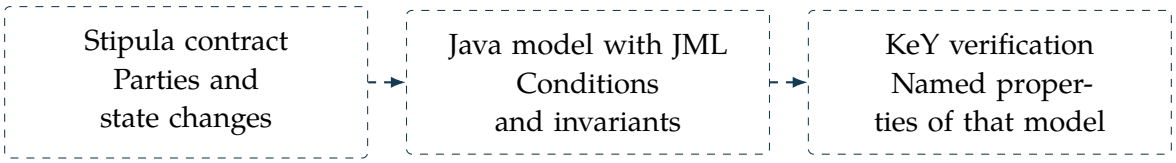


RuleIR is one target of the shared model. Native Catala does not need to pass through RuleIR. A different proposed meaning requires a different model, not merely a different target.

For example, a gift restriction and a separately specified disclosure duty can both belong in a package. Two rival readings of the restriction remain alternative packages. The proposed grouping and each unresolved question are preserved. There are two Catala routes. The compatibility route lowers a supported RuleIR profile and retains the existing result, trace and release checks. The native route consumes the shared model directly, so rich records, lists, options, payload variants and typed operations can be reported as executable Catala even when the RuleIR target returns an explicit capability failure. These routes share the frontend and evidence policy; they do not make different interpretations agree by construction.

5 continued. Where Stipula belongs

Stipula describes contracts through parties, resources, actions and state changes. It is relevant when consent can be granted and withdrawn. The published translation studied in Chapter 4 constructs a Java verification model with specifications in the Java Modeling Language (JML). KeY is a verifier that reasons about the Java model and those specifications.

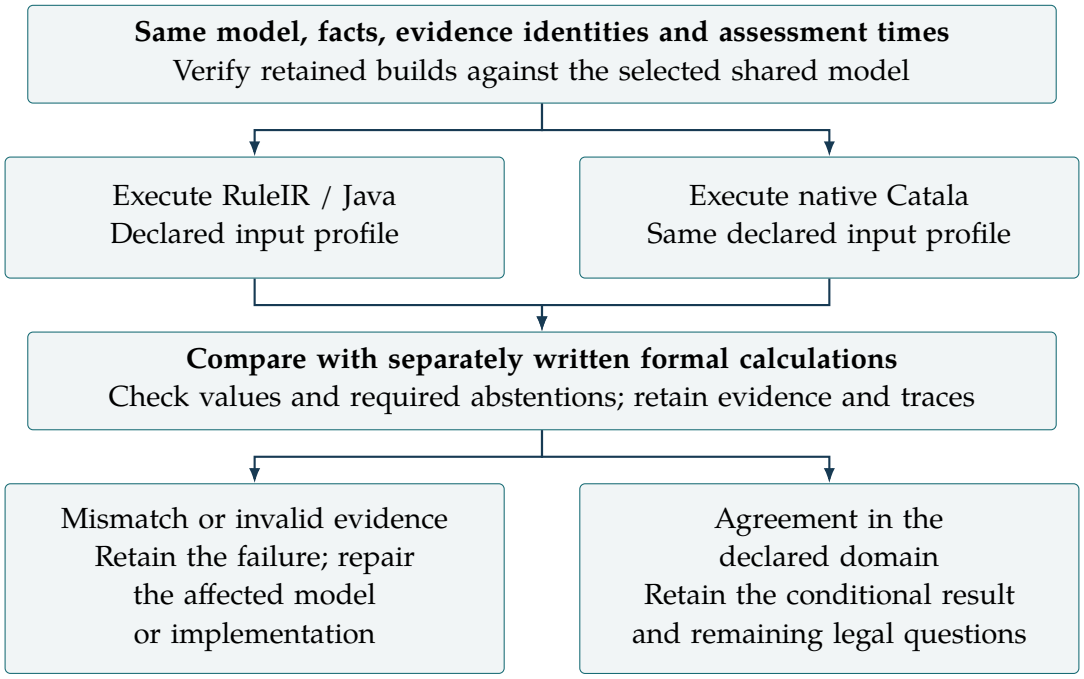


Dashed boxes show a studied research route, not the current LegalMath compiler. Its Java model is not automatically a bank application.

The retained translator examples contain a compilation discrepancy; no local KeY replay establishes their reported proofs. Adoption requires reproducible inputs and a justified mapping to LegalMath's event rules. Chapter 4 details those limits; Chapter 7 explains the implemented rule language.

6. Generate executable targets and check their behaviour

The Java emitter embeds the fixed RuleIR rules in a generated policy class that calls a shared Java evaluator. The Java compiler, `javac`, compiles that class and its supporting runtime. A separately written Python evaluator supplies another execution of the same proposed rule. The Catala translators take the same shared model when their capability report permits it: the compatibility route lowers supported RuleIR, while the native route emits richer Catala directly. Each target receives the same declared facts and assessment times for the cases in its profile.



The Java/Python pair and the Catala routes check whether a proposed rule survives translation within a declared profile. They can agree on a rule that omits a legal exception.

The bank and CoCo runs use a complete-input profile: unknown, conflicting or expired required evidence causes abstention. In the paired run, each target returned 138 complete-input results and 19 abstentions. A successful abstention test demonstrates preserved uncertainty, not a favourable purchaser finding.

The test cases must also be capable of exposing mistakes. Boundary cases place values on either side of a threshold. Missing-fact cases check that uncertainty survives. Event histories exercise consent, withdrawal and timing. Deliberately altered rules, called *mutations*, ask whether the tests notice a removed exception or a reversed condition.

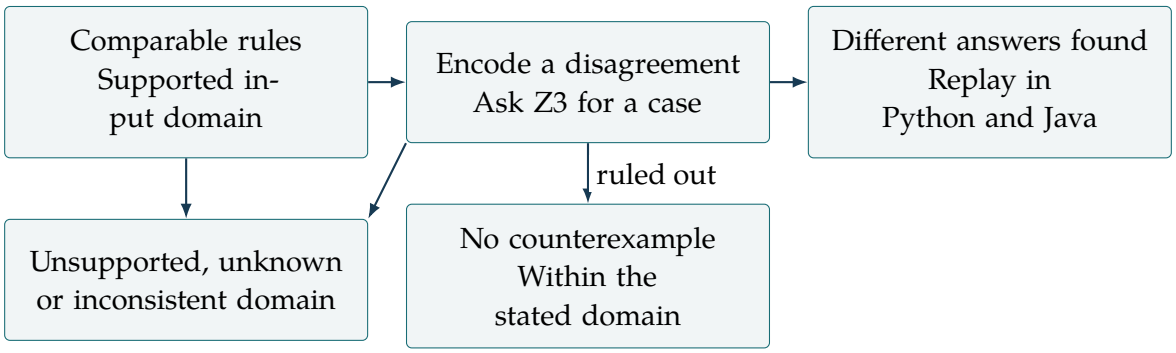
For a small finite rule, every declared input combination can be enumerated. Larger or unsupported domains require more limited conclusions. Expected legal outcomes need their own source justification, separate from Python/Java agreement. Compilation failures and disagreements return to the affected rule or implementation; successful checks accompany the draft rather than remove its open interpretation questions. Chapter 7 develops these checks and the separate Java host demonstration.

6 continued. Where mathematical verification fits

A mathematical check needs a precise proposition: whether two encoded rules differ on a permitted input, for example, or whether a transfer preserves a quantity. Such propositions concern the formal model and its assumptions.

Symbolic comparison with Z3

Z3 is a satisfiability modulo theories (SMT) solver: it searches for values that make a collection of logical and arithmetic conditions true. Here the condition asks whether two supported rule encodings return different answers. The facts, output meanings and permitted input domain must first be made comparable.



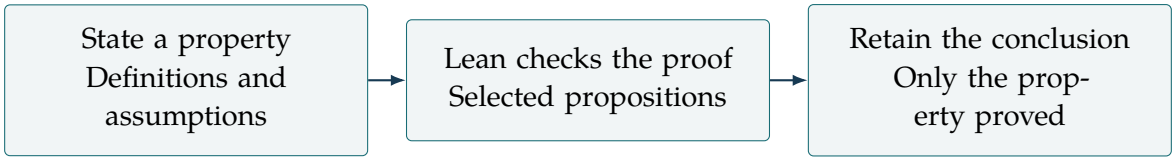
A solver result concerns an encoded comparison. An unsupported operation, timeout or empty domain cannot count as agreement.

The solver adapter covers a stated subset of the formal model and does not check an independent proof certificate from the solver. A found discrepancy is valuable because the original programs can replay it. A no-counterexample result retains the encoding and domain assumptions. Unsupported operations, timeouts and empty domains remain explicit outcomes. Chapter 3 explains those restrictions.

6 continued. Prove a translation and challenge the tests

Proofs of selected mathematical properties

Lean is a proof assistant that checks a formal derivation. The mathematical review used it for selected logical and arithmetic propositions, including three-valued logic, an exception counterexample and conservation under a stated transfer. SymPy, a symbolic algebra system, checked selected algebraic identities.



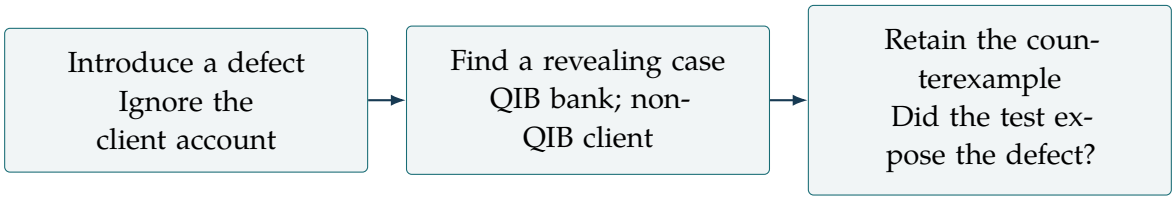
These are checks of selected mathematical claims, not an automatic proof of each generated program or of the whole compiler.

The successor also registers a narrow Lean certificate for Boolean expression lowering. Its fixed checker proves the relation between two formal descriptions; an additional finite check executes the generated Java for every declared Boolean state assignment within its size limit. Arithmetic, dates, full event histories and English meaning remain outside that theorem. A separate Java entry point demonstrates attributed contact histories, keeping the duty trigger distinct from an observed contact and from overall compliance.

For the CoCo purchaser rule, a Lean certificate checks preservation from the formal expression to the shared model. The independent solver comparison covers all 512 assignments of the nine Boolean premises. Native execution is a separate check: both targets receive the same 157 complete, uncertain or expired cases. These are different forms of evidence about a stated model.

Would the tests detect a consequential mistake?

The purchaser tests deliberately remove conditions that matter. One mutation ignores whose account benefits; another ignores the purchaser's US-person status. A revealing input must make the altered rule disagree with the independently expressed relation.

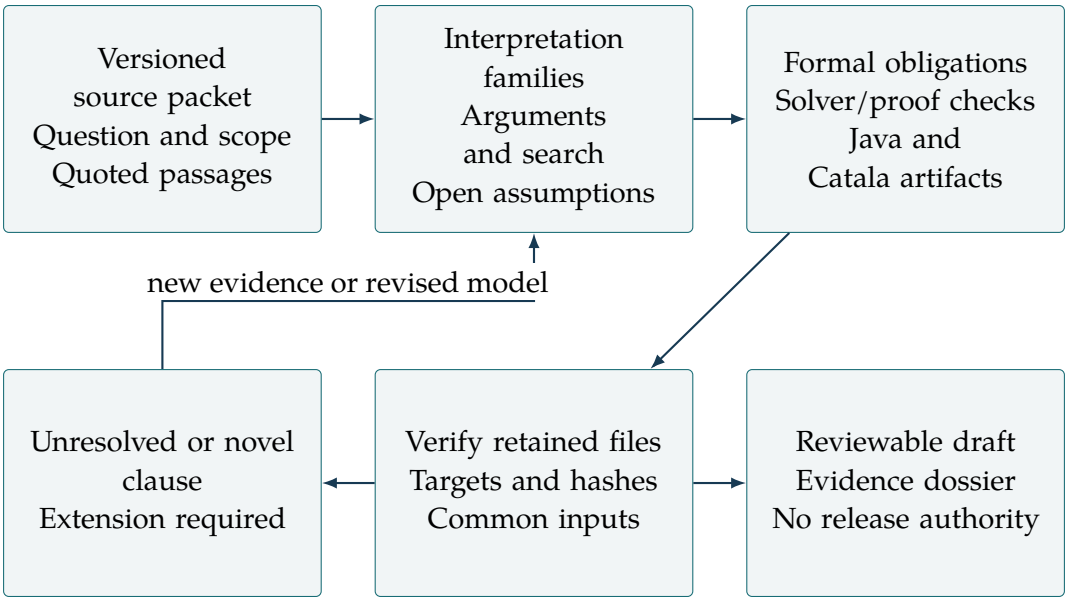


Six altered purchaser rules produced counterexamples. Mutation testing checks sensitivity to those defects; it does not manufacture legal answers or establish that the source interpretation is complete.

Proving that every supported rule preserves its behaviour through translation remains a stronger goal. The present checks do not establish that whole-compiler result or correct interpretation of English. The Stipula/JML/KeY route offers another research approach to particular Java properties.

The proof-carrying dossier and the abstention path

The preceding checks are useful only when their targets and assumptions stay visible. The integration record therefore binds a source packet, competing hypotheses, arguments, proof obligations, method-family results and executable artifacts by hash. It also records shared dependencies: agreement between Java and Catala after the same RuleIR lowering is a translation check, not two independent readings. The controller never changes an unavailable, unknown or unsupported result into agreement.



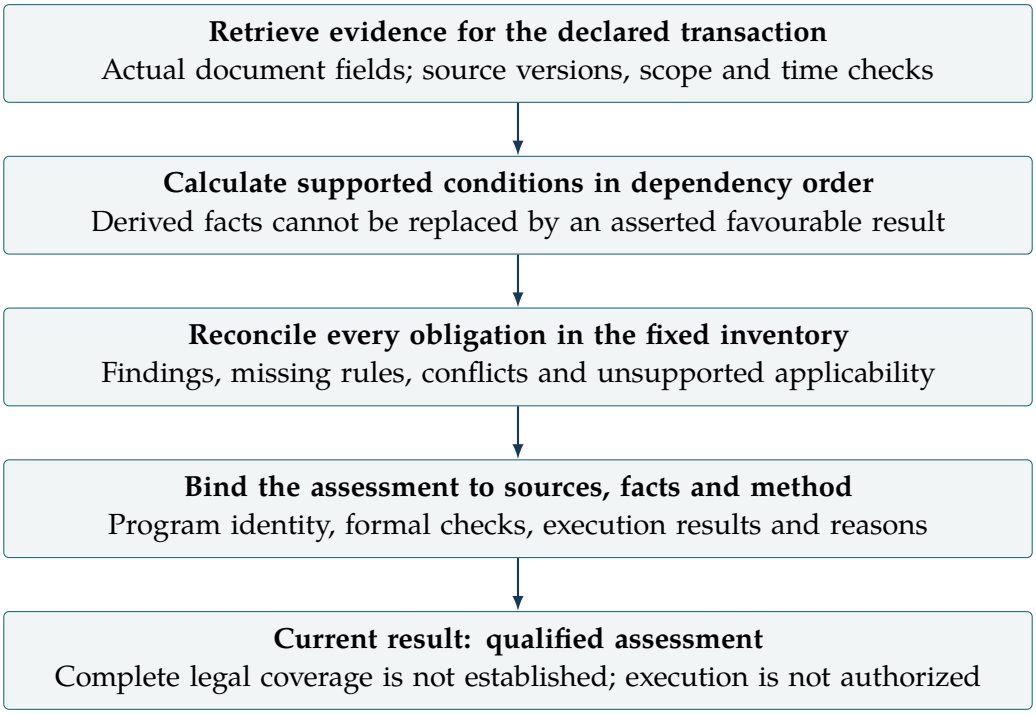
A proof or replay closes the proposition it names. An open authority, future clause, stale result or unavailable required method keeps the dossier on the abstention path.

The gift-circular replay (23EC46) imports five previously proposed readings. Two differ on whether product-directed presentation suffices without an operational connection. The solver finds a separating assignment and the original Python and Java programs replay it. The Catala routes supply another execution target over the shared model: the compatibility backend consumes supported RuleIR, while the native converter can consume a richer model without first making it RuleIR. Agreement after a common lowering is a translation check, not independent legal evidence. The disabled live provider supplies no fresh reading. The witness identifies a legal question; it does not answer it. The result can be delivered as a draft Java package with its evidence and open questions. The transaction assessment must retain the unresolved classification and any missing policy. The exact contract and master command are described in Section 6.34.

The master validates a frozen revision and binds each accepted phase to its inputs and predecessor files. Failure preserves the attempt and triggers a focused repair check before a bounded retry. Success refreshes the next-phase plan. A changed source, checker, plan or allowlist invalidates acceptance; retry exhaustion produces a report of the remaining failure. Neither a retry nor a successful document build supplies missing interpretive evidence.

7. Assemble the transaction assessment

A Java archive (JAR) or a declared Catala package contains the compiled control. Its execution result is only one part of the transaction assessment. The investigation also checks the fixed obligation inventory, applicability, retrieved factual documents and their source dependencies. Each finding keeps its scope, supporting evidence and unresolved questions.



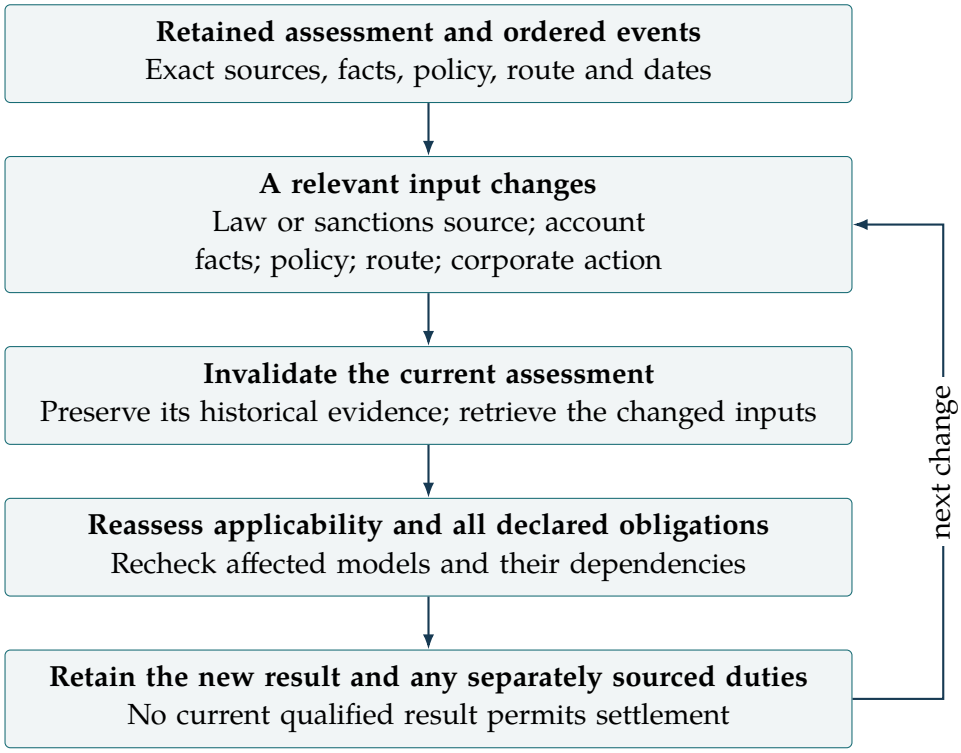
A request supplies evidence and context, not its own shorter list of requirements or precomputed favourable findings. The retained assessment can be reconstructed from its bound inputs.

For example, account amounts enter the financial condition before client qualification is calculated. A transaction threshold also needs its separately declared controls premise. Sophisticated professional investor eligibility then feeds the condition for streamlined procedures. A false sufficient condition does not prove its converse: failure to establish that a bond is complex does not establish that it is non-complex.

The CoCo purchaser result is supplementary evidence attached to this assessment. The original fourteen bank obligations remain present for each instrument. Actual JPMorgan entity, account, policy and settlement facts have not been supplied. The program therefore retains those gaps and returns no transaction execution permission. Recording a bank authorization as a business fact would not turn it into proof of legal interpretation quality.

8. Reassess when evidence or events change

An assessment belongs to particular source editions, factual records, a method and an assessment time. It cannot be reused as though a later transaction had the same circumstances. The event replay records changes and invalidates the current assessment before it considers a later settlement request.



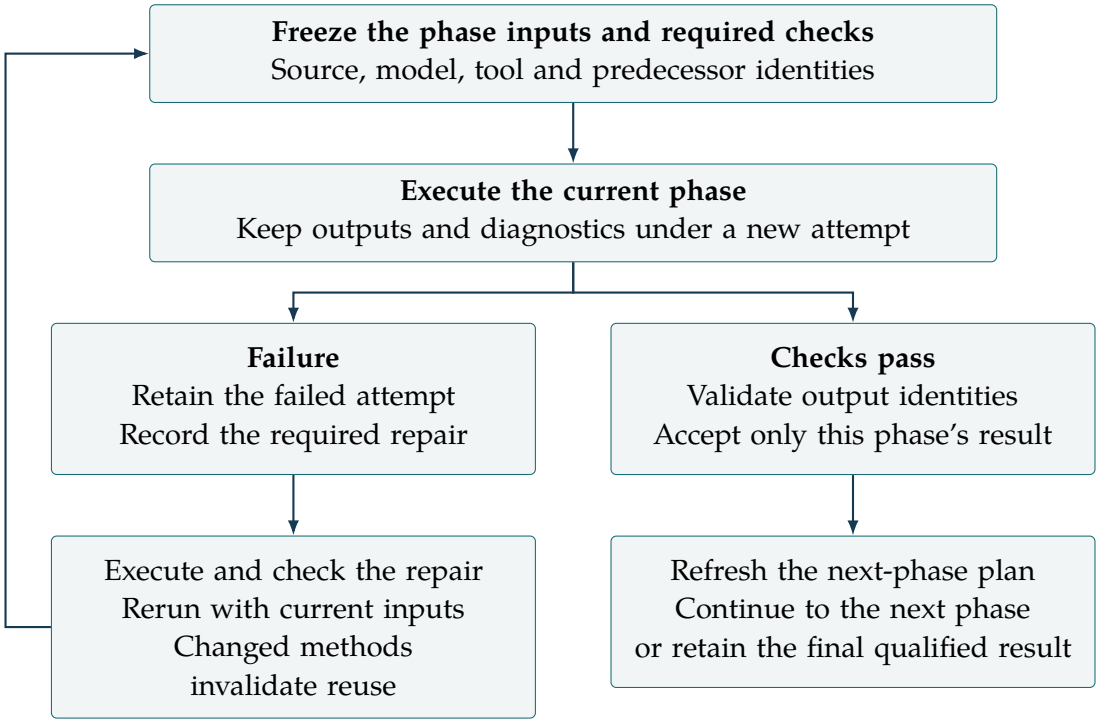
Settlement requires a current assessment for its declared time. A stale or absent assessment is rejected. Passing that freshness check still does not turn a qualified result into permission to trade.

Duties keep the responsible actor, trigger, source and due time separate. Rejection, blocking assets, reporting, rescreening and disclosure are different actions; a sanctions finding cannot silently substitute one for another. A business-day deadline needs an explicit retained calendar within its supported range. Evidence recorded as completion does not prove the action occurred.

Official sanctions snapshots can be refreshed and checked before replacement; earlier bytes remain available for historical investigations. Exact-name matches identify candidates for entity resolution. Ownership propagation for a blocking program cannot automatically be reused for a solely Chinese military-industrial company designation. These scoped mechanisms do not complete identity resolution or every sanctions program.

Repair the implementation; refresh the next phase

The executable master programs preserve the evidence from each phase. The bank and paired-CoCo campaigns validate sources and tools, run regressions, check formal relations, execute both targets, and then assemble the integrated investigation. A phase failure must remain visible until a repair has been executed and checked.



The runner records a repair request; a failed check does not repair itself. Reuse of a successful phase requires matching inputs and output hashes. An engineering pass does not remove the legal qualifications.

Evaluate new situations without human answer labels

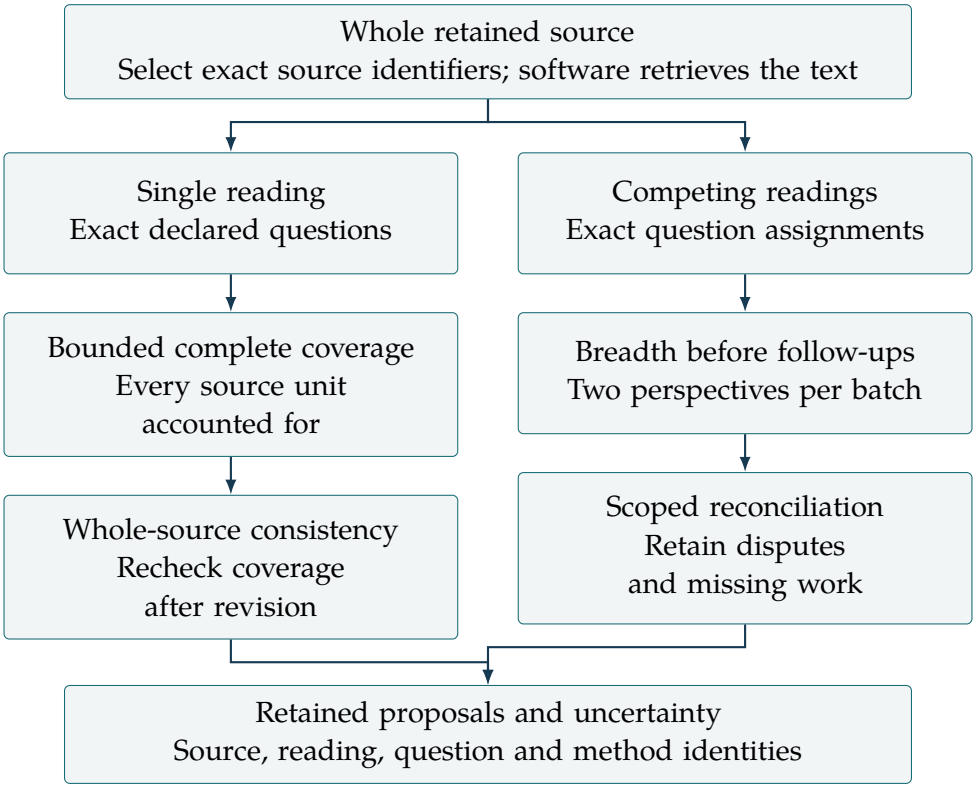
The product requires a machine-checked proposition or an explicit qualification. Human approval and agreement among models are not quality evidence. Independent equations, translation proofs, adversarial cases and source-bound challenges test different possible failures, each within its declared domain.

For future-source work, the observation mechanism freezes the method, inventory and prior sources before it accepts new content. A changed method starts a new series. New-to-inventory documents and generated amendments test novelty handling; they do not establish accuracy on future laws. The present bank series has no real prospective observations. Unknown meaning must remain qualified until evidence establishes the particular claim being made.

Complete the processing without losing the question

The same proof-or-qualification requirement applies when a source investigation must be split into smaller pieces. Finishing those pieces matters because an unexamined exception can invalidate a conclusion drawn from the rest.

Long responses and exhausted review budgets can leave parts of a circular unexamined. Two implemented paths share exact source references. The single reader accounts for its declared questions; the ensemble investigates competing readings. These paths retain their separate results. See Section 6.36 for the mechanics and checked limits.

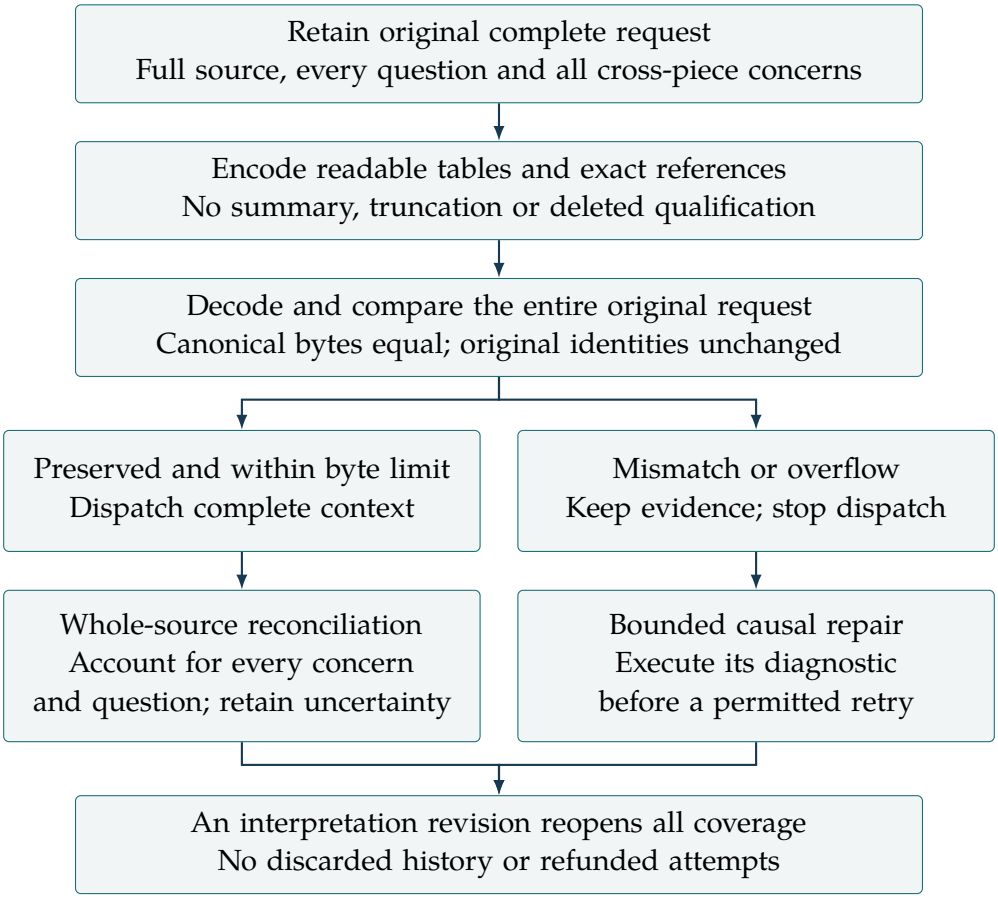


An interpretation revision reopens its coverage checks. Failed requests keep their spent attempts; no batch disappears to fit the budget.

The qualification adapter checks supported formal propositions for a retained reading and exact question. The complete ensemble investigation invokes it when enabled; the split single-reader path retains proposals for subsequent checking. A source identifier proves where the words came from. Coverage proves that the declared units and questions received dispositions. A formal theorem proves its stated relation. None of these alone proves the English interpretation. The four earlier unfamiliar investigations therefore remain incomplete unless their remaining stages actually execute under the recorded limits.

Preserve every concern before shortening a request

The retained authentication exercise reached all 263 source units but could not fit its final whole-source comparison into one request. The optional readable table encoding removes repeated representation while retaining the complete source, questions and explanations. Section 6.37 explains the encoding and its executed checks.

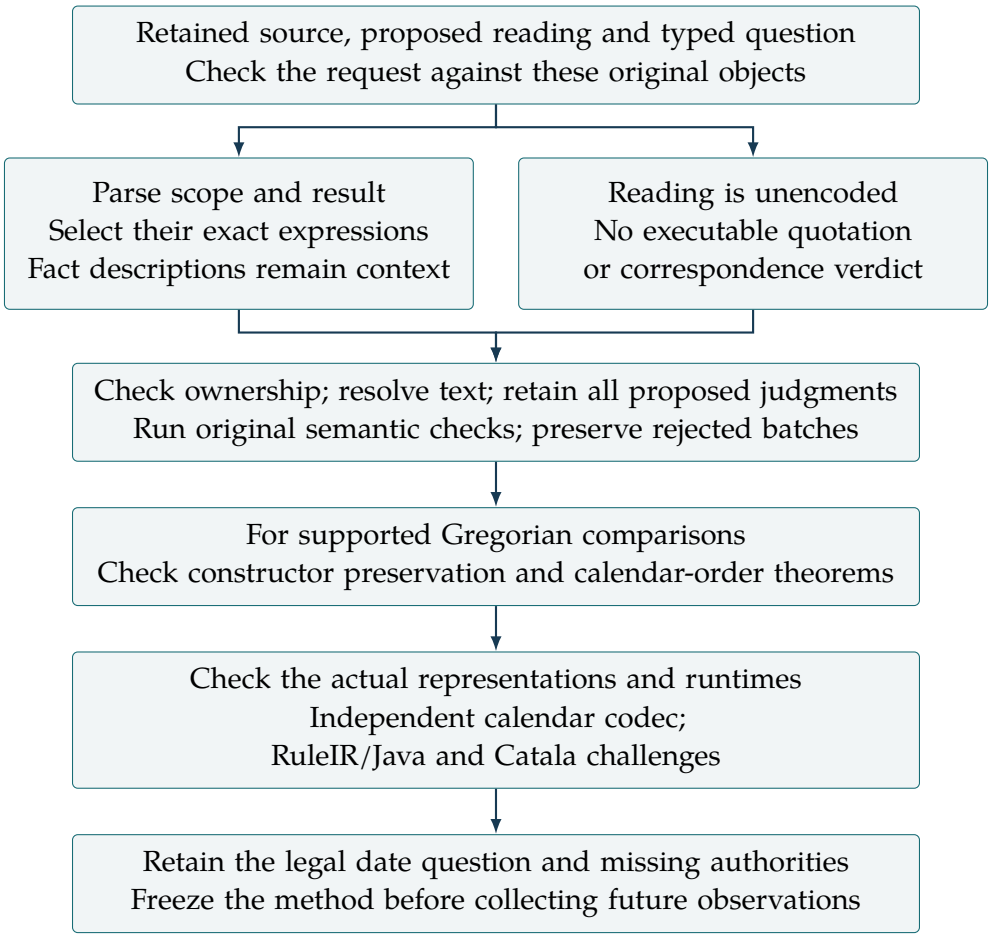


Exact reconstruction permits the capacity comparison; it does not establish legal support. The executed synthetic reconciliation retained every concern as unresolved.

The measured request shrank from 382,396 to 175,016 bytes under the unchanged 200,000-byte limit. A challenge with long unique explanations still exceeded that limit and stopped before dispatch. A future source can therefore produce an explicit capacity qualification. Model comprehension of the encoding and English correctness remain unestablished by these offline exercises.

Check program references and dates independently

Section 6.38 separates exact quotation, mathematical preservation and native execution. Each establishes a different proposition.



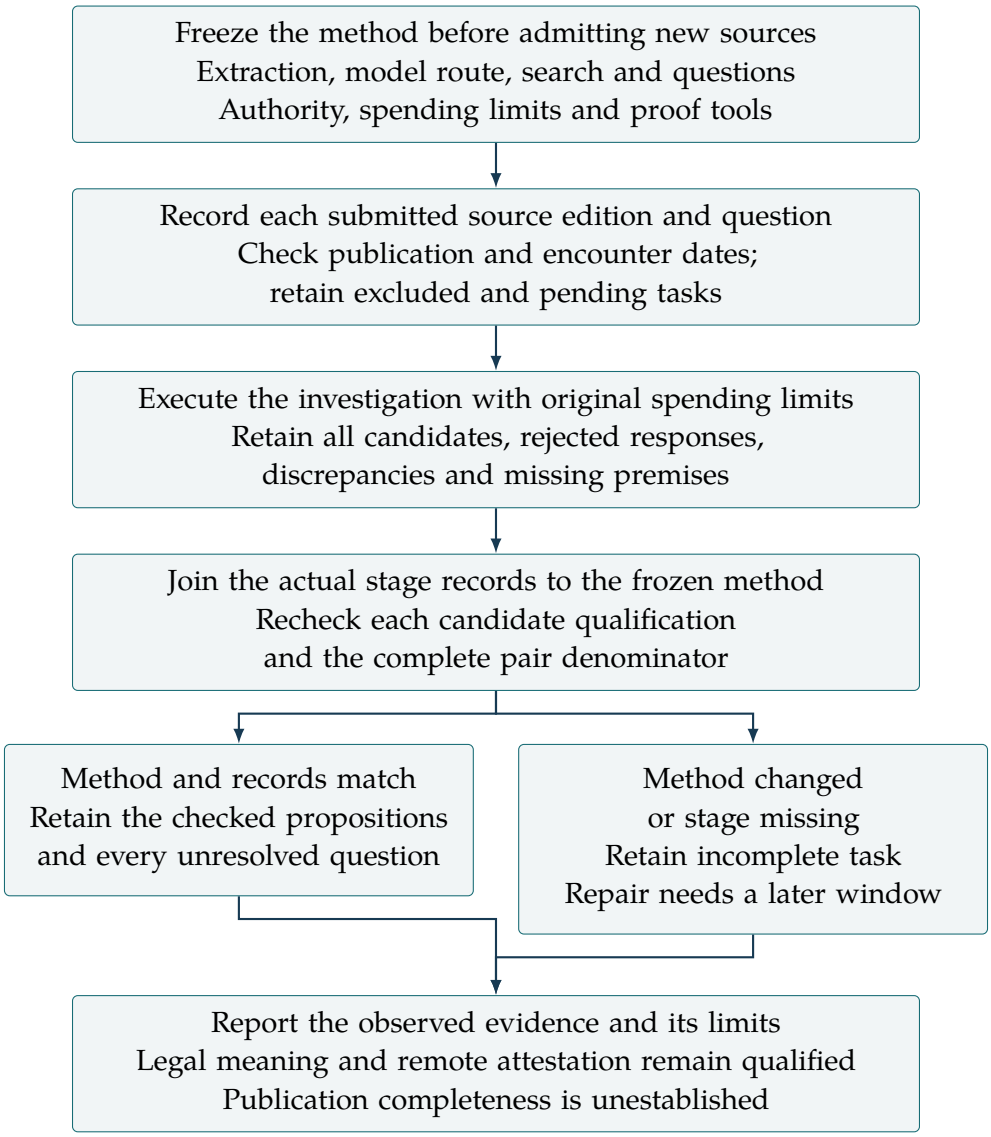
The calendar domain is valid proleptic Gregorian dates in years 1–9999. Business-day rules and selection of the legally relevant date remain separate.

The eight encoded UCITS readings can now receive date-preservation evidence. The ninth remains unencoded, and unresolved source questions persist for all nine. Tests of this implementation do not create future observations or certify the interpretation of an unknown future legal text.

Source extraction and question definition use source references. Version 2 executable references apply only at the correspondence stage. A failed response triggers bounded diagnostic processing; exhaustion retains every unresolved pair and concern. Exact quotation never establishes legal support.

Keep the method fixed when observing future sources

Section 6.38.7 explains why a study must identify the extraction and interpretation process as well as its final proof checker.

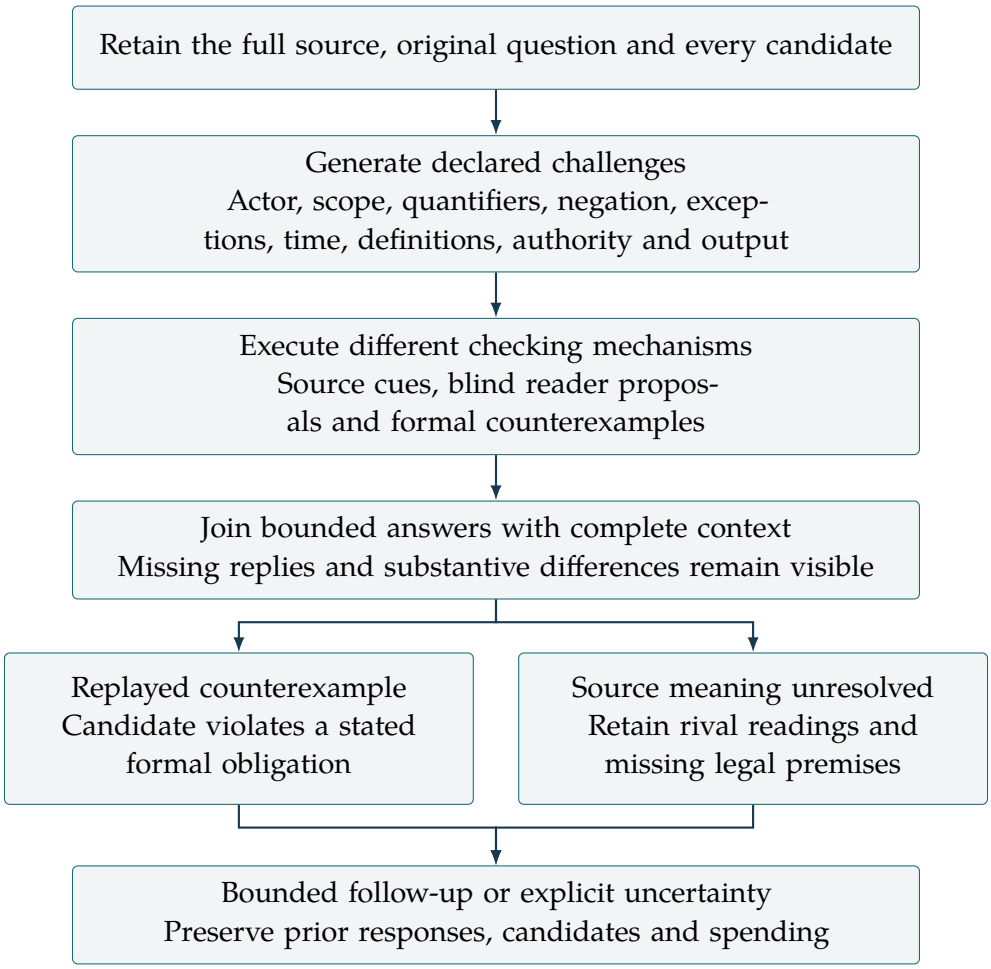


A checked program belongs to a particular investigation. The same program produced from a changed source or method is a different observation.

An additional allowance does not reset the original question's attempts or deadline. A revised source keeps its predecessor's unresolved questions. A repaired method needs a later untouched observation window; no earlier failure is silently turned into evidence for that revision.

Challenge a reading before eliminating it

Section 6.39 explains the distinction between an unresolved source reading and a checked mathematical counterexample.



A counterexample rejects a candidate relative to a declared premise. It does not establish that the English source warrants that premise.

The source scanner cannot prove that all exceptions were found. A finite grammar cannot prove that every legal interpretation was considered. These qualifications remain explicit when the checked stages are joined into one investigation.